

2023

YEAR 11 YEARLY EXAMINATION

Mathematics Standard

Examiner: Sami El Hosri

General Instructions

- Reading Time – 10 minutes
- Working Time – 2 hours
- Write using black pen
- Calculators approved by NESa may be used
- A separate reference sheet is provided at the back of this paper.
- All necessary working should be shown in Questions 16 – 36

Total marks – 75

Section I: Multiple Choice

Questions 1 – 15 15 marks

- Attempt all questions
- Allow about 25 minutes for this section

Section II: Extended Response

Questions 16 – 36 60 marks

- Attempt all questions
- Allow about 1 hour 35 minutes for this section

Section I

Questions 1 – 15 (15 marks)

Read each question and choose an answer A, B, C or D.

Allow about 25 minutes for this section.

Use the multiple-choice answer sheet for Questions 1-15.

- 1 Which expression will give the area of a circle sector with an angle of 117° , given that the diameter is 12 cm ?

A) $\frac{117}{360} \times \pi \times 6^2 \text{ cm}^2$

B) $\frac{117}{360} \times \pi \times 24^2 \text{ cm}^2$

C) $\frac{117}{360} \times \pi \times 12 \text{ cm}^2$

D) $\frac{117}{360} \times \pi \times 24^2 \text{ cm}^2$

- 2 The sum of the scores of a set of four scores is 30.
The sum of the lowest score and the highest score is 16.
What is the median of these four scores?

A) 14

B) 8

C) 7

D) 5

- 3 The length of a park is measured to be 424.8 m.
What is the percentage error in this measurement correct to four decimal places?

A) $\pm 0.1177\%$

B) $\pm 0.0117\%$

C) $\pm 0.1178\%$

D) $\pm 0.0118\%$

- 4 In which of the following linear equations will the value of y decrease by 2 when the value of x is increased by 3?

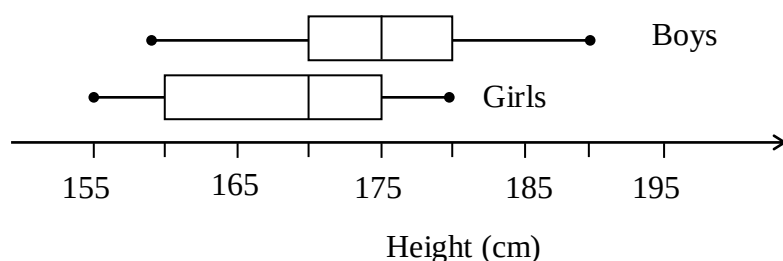
A) $y = -\frac{2}{3}x$

B) $y = -\frac{3}{2}x$

C) $y - 2 = x + 3$

D) $y - 3 = x + 2$

- 5 In a Year 11 class there are 16 boys and 14 girls. The box and whisker plots below show the distributions of their heights.



How many students in this class are at least 170 cm tall?

- A) 7
- B) 12
- C) 15
- D) 19
- 6 A box contains 12 apples. 7 of these apples are green and 5 are red. Amanda is to select 2 apples from the box at random. What is the probability that Amanda selects two red apples?

A) $\frac{5}{12} + \frac{5}{11}$

B) $\frac{5}{12} \times \frac{4}{11}$

C) $\frac{5}{12} + \frac{4}{11}$

D) $\frac{5}{12} \times \frac{4}{12}$

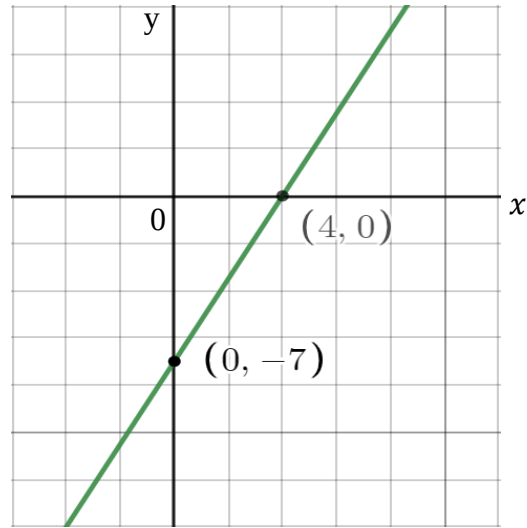
- 7 The graph shows a line (ℓ).
Which of the following is the equation of line (ℓ)?

A) $y = \frac{7}{4}x - 7$

B) $y = \frac{7}{4}x + 4$

C) $y = -\frac{7}{4}x - 7$

D) $y = \frac{7}{4}x - 4$



- 8 Last year, Kaydon bought 800 shares at \$4.40 per share. This year the price of each share is \$6.25. Kaydon receives a dividend of \$1.20 per share.

What is the dividend yield of these shares?

A) $27\frac{3}{11}\%$

B) $\frac{3}{88}\%$

C) 19.2%

D) 0.024%

- 9 In a world cup match, 90 000 people were at a stadium to watch a game. At the end of this game they started to leave at the rate of 4500 people per minute.

Which of the following equations shows the number of people N that are still in the stadium p minutes after the game finished?

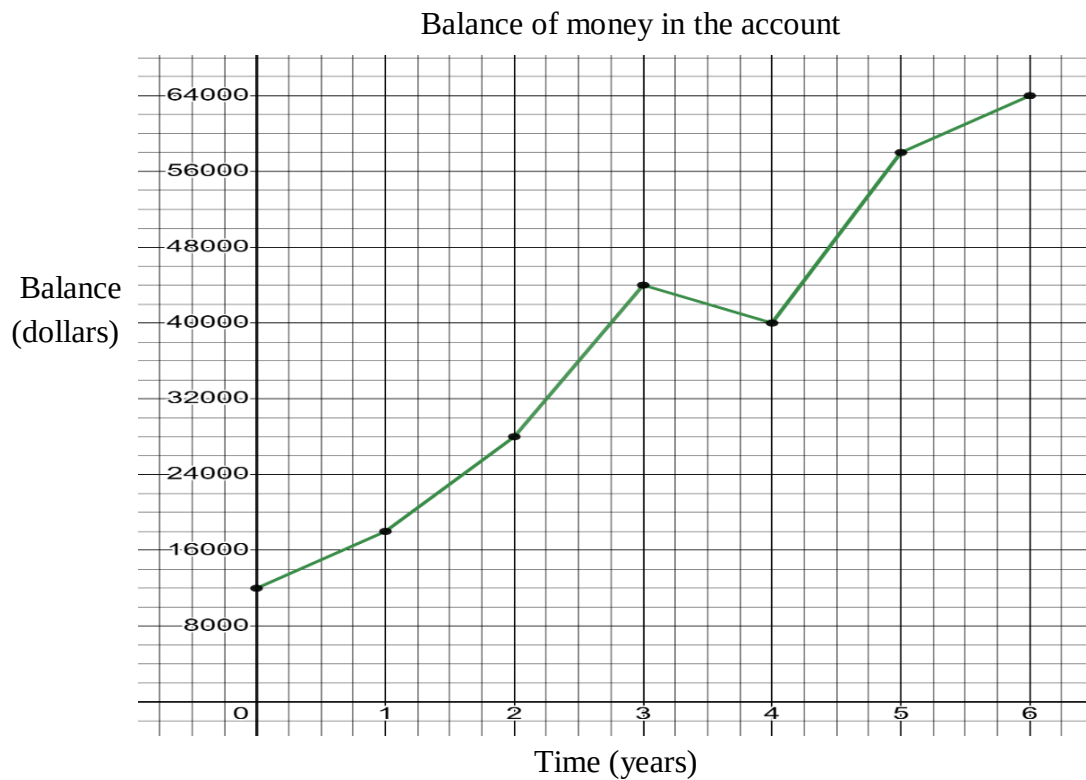
A) $N = 4\,500p + 90\,000$

B) $N = 4\,500p - 90\,000$

C) $N = 90\,000p - 4\,500$

D) $N = 90\,000 - 4\,500p$

- 10 The graph below shows the changes in the balance in Tania's saving account over the last six years.



- What is the average percentage increase in the balance per year over these six years, to the nearest whole percent?
- A) 14%
- B) 72%
- C) 81%
- D) 433%
- 11 The volume of a hemisphere is $486\pi \text{ cm}^3$.
- What is the curved surface area of this hemisphere?
- A) $324\pi \text{ cm}^2$
- B) $204\pi \text{ cm}^2$
- C) $162\pi \text{ cm}^2$
- D) $102\pi \text{ cm}^2$

- 12 The dosage of medication for children aged 1 to 12, based on an adult dosage, can be calculated using Young's formula shown below.

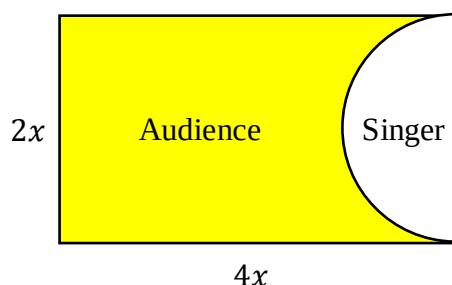
$$\text{Child dosage} = \frac{\text{age of child (in years)} \times \text{adult dosage}}{\text{age of child (in years)} + 12}$$

Rachel is 8 years old and her sister Clara is 4 years old. Both are sick and need to take different dosages of the same medicine.

According to Young's formula, Rachel needs a dosage of 9.6 mL.

What is the dosage that Clara needs?

- A) 4.8 mL
- B) 6.0 mL
- C) 6.4 mL
- D) 5.6 mL
- 13 A concert hall is to be built in the shape of a rectangle with width $2x$ metres and length $4x$ metres.



The hall is to be divided into two parts. One part is a semicircle for the singer and the remaining part of the rectangular hall is for the audience.

What is the area of the audience section?

- A) $8x^2 - \frac{1}{2}\pi x^2$
- B) $8x - \frac{1}{2}\pi x^2$
- C) $8x^2 - \pi x^2$
- D) $\frac{1}{2}\pi x^2 - 8x^2$

- 14** Kevin's normal hourly rate is \$21.50. Last week he worked certain hours at normal time, and some hours at time and a half.
- The number of hours he worked on normal time were three times the number of hours he worked on time and a half.
- His total pay for the week was \$774.
- How many hours did he work in total last week?
- A) 16 hours
- B) 24 hours
- C) 28 hours
- D) 32 hours
- 15** Layla's tax payable for this financial year is calculated using the tax table shown below.

Taxable income	Tax on this income
0 – \$18,200	Nil
\$18,201 – \$45,000	19 cents for each \$1 over \$18,200
\$45,001 – \$120,000	\$5,092 plus 32.5 cents for each \$1 over \$45,000
\$120,001 – \$180,000	\$29,467 plus 37 cents for each \$1 over \$120,000
\$180,001 and over	\$51,667 plus 45 cents for each \$1 over \$180,000

Layla's taxable income this financial year increases from \$110 000 to \$135 000.

What percentage of this increase in her income will she keep after paying the additional tax?

- A) 62%
- B) 64.8%
- C) 65.25%
- D) 63%

Year11 Mathematics Standard

Section II Answer Booklet

60 marks

Attempt Questions 16 – 36

Allow about 1 hour and 35 minutes for this section.

INSTRUCTIONS

- Answer the questions in the spaces provided. Sufficient space is provided for typical responses.
 - Your responses should include relevant mathematical reasoning and/or calculations.
 - Extra writing space is provided at the back of this booklet.
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QUESTION 16 (2 marks)

Kathy surveyed six people. She asked them how much they paid for water in their last bill.

Here are the results: \$250, \$280, \$305, \$220, \$420, \$367.

a) Calculate the mean of this set of data.

1

b) What is the standard deviation of this set of data, correct to two decimal places?

1

QUESTION 17 (2 marks)

Draw a possible graph for a positively skewed distribution curve showing the possible location of the median and the mode.

2

QUESTION 18 (2 marks)

A TV was purchased 2 years ago for \$2 550. Its value depreciates by 20% every six months.

2

What is the current value of this TV?

QUESTION 19 (2 marks)

Gavin is to survey the employees in the four branches of his company. He decides to use a stratified sample.

2

The total number of employees in these four branches of his company is 320 employees. There are 120 employees in the main branch.

If Gavin decides on a sample group of 40, how many employees from the main branch should be surveyed?

QUESTION 20 (2 marks)

A rectangular piece of wood is measured and is found to be 48 cm long and 24 cm wide.

2

What are the upper and lower limits of the area of this rectangle ?

QUESTION 21 (2 marks)

Joanne's fortnightly wage is \$2 460. She decides to take her four weeks annual holiday.

2

Find her holiday pay, which consists of her normal pay for the four weeks plus 17.5% holiday loading.

QUESTION 22 (2 marks)

It takes Jared 36 minutes once every 5 days to iron his clothes.

2

The iron uses 2.7 kW. The electricity is charged at 27.5 cents/kWh when the iron is in use.

What is the cost of electricity that Jared uses every 40 days for ironing?

QUESTION 23 (3 marks)

Kate is a sales representative. She receives a basic weekly wage of \$420 and a commission of 8.5% on whatever she sells over \$3 200 in the week.

3

If this week she sells \$12 100 worth of goods, what will be her weekly wage?

QUESTION 24 (3 marks)

Ricardo invests \$80 000 for 4 years at 3% per annum compounded monthly.

3

Find the total amount of interest Ricardo will earn on his investment over the 4 years.

QUESTION 25 (3 marks)

Data was collected from 20 people on the number of messages that they made in the last month. The set of data collected is displayed in the stem and leaf plot.

3

Messages

1		0
2		0 4
3		0 0 0 2 5 5 6
4		0 0 0 0 2 2 2
5		0 0
6		4

Is 64 an outlier for this set of data? Justify your answer using suitable calculations.

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QUESTION 26 (2 marks)

City A and City B have Coordinated Universal Time of -10.00 and $+4.00$ respectively.

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Jane flew from City A on a Friday at 4:20 am to go to City B.

The flight took 12 hours.

On what day and at what time did Jane's plane arrive at City B?

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QUESTION 27 (5 marks)

Ronald weighs 74 kg. He ate 176 g of salmon and 48 g of pasta.

The labels on these products indicate that the salmon has 220 kJ of energy in a 55 g serve, and the pasta has 1020 kJ in a 120 g serve.

- a) Calculate the number of kJ in his salmon and pasta.

3

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- b) The table below contains information indicating the number of kilocalories used each minute in a variety of exercises for different body masses.

2

Activity	56 kg	68 kg	74 kg
Volleyball	2.8	3.4	3.7
Dancing	7.5	9.2	10.0
Tennis	6.1	7.4	8.1

1 kilocalorie is equivalent to 4.184 kilojoules

How many minutes must Ronald play tennis in order to use all the energy in the salmon and the pasta? (Answer to the nearest minute.)

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Question 28 (4 marks)

On Rocky Beach, the cost C in dollars of hiring a speed boat from Boat Champions is given by

$$C = 65.6 t + 98.4, \text{ where } t \text{ is the time in hours.}$$

- a) What is the cost to hire a speed boat for 4 hours and 15 minutes from Boat Champions?

1

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- b) The cost of hiring a speed boat from Boat Kingdom is given by
 $C = 72.3 t + 78.6$, where t is the time in hours.

3

Last Friday, Tania saved \$13.70 by hiring a speed boat from Boat Champions instead of hiring it from Boat Kingdom.

For how many hours did Tania hire the speed boat last Friday?

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Question 29 (3 marks)

Jennifer bought a car with a cash price of \$32 000 on terms.

3

She agreed to pay 40% deposit and to pay the remaining amount plus interest over two years in equal instalments.

If simple interest at the rate of 4% p.a. was charged on the remaining amount, with six months interest free, how much did Jennifer pay each month?

Question 30 (3 marks)

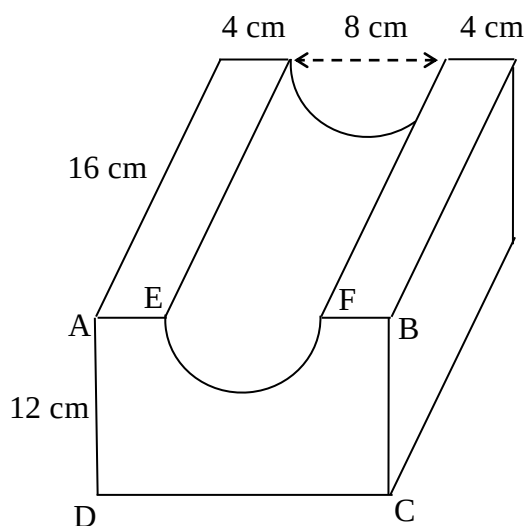
The intensity I of a current varies inversely with the resistance R in a circuit. The intensity $I = 24$ when $R = 6$.

Find the intensity I when $R = 8$.

3

Question 31 (4 marks)

The cross section of the prism shown in the diagram consists of a semi-circle with diameter EF removed from a rectangle ABCD.



a) Find the area of the cross section correct to four decimal places.

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(b) Hence, find the volume of the prism correct to four decimal places.

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Question 32 (3 marks)

This year the income tax rate for a taxable income over \$120 000 and up to \$180 000 is as shown in the table below.

Taxable income	Tax on this income
\$120 001 – \$180 000	\$29 467 plus 37c for each \$1 over \$120 000

Matthew's gross annual salary is \$175 000 and his allowable tax deductions for the year are equivalent to 9% of his gross income.

What is Matthew's tax payable on his taxable income for this year?

3

Question 33 (3 marks)

In a game Patricia is to roll a normal pair of dice twice. She wins a major prize if the sum of the numbers shown on the uppermost faces is at least 10 on each of these two rolls.

What is the probability that Patricia wins a major prize?

3

4

He stopped drinking at 11:20 pm and left the party at 2:50 am just when his blood alcohol content (BAC) reached zero.

$$\text{using Number of hours for BAC to reach zero} = \frac{\text{BAC}}{0.015}$$

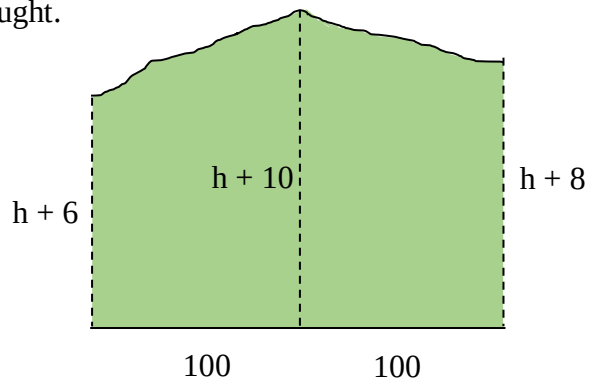
You will also need the formula

$$BAC_{male} = \frac{10N - 7.5H}{6.8M}$$

This image shows a single sheet of white paper with horizontal ruling lines. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

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The area of this land using two applications of the trapezoidal rule area is about 6100 m^2 .



Find the value of h.

3

This image shows a full page of white paper with horizontal dashed lines, typical of primary school writing paper. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

QUESTION 36 (3 marks)

The ratio of the number of black cars to white cars at a car dealership was 5:2.

When 10 black cars were sold, and 14 white cars added to their fleet, the ratio became 8:5.

How many black cars did the car dealership have at first?

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Section II extra writing space

If you use this space, clearly indicate which question you are answering.

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Mathematics Standard 1

Mathematics Standard 2

REFERENCE SHEET

Measurement

Limits of accuracy

Absolute error = $\frac{1}{2} \times \text{precision}$

Upper bound = measurement + absolute error

Lower bound = measurement – absolute error

Length

$$l = \frac{\theta}{360} \times 2\pi r$$

Area

$$A = \frac{\theta}{360} \times \pi r^2$$

$$A = \frac{h}{2}(a + b)$$

$$A \approx \frac{h}{2}(d_f + d_l)$$

Surface area

$$A = 2\pi r^2 + 2\pi rh$$

$$A = 4\pi r^2$$

Volume

$$V = \frac{1}{3}Ah$$

$$V = \frac{4}{3}\pi r^3$$

Trigonometry

$$\sin A = \frac{\text{opp}}{\text{hyp}}, \quad \cos A = \frac{\text{adj}}{\text{hyp}}, \quad \tan A = \frac{\text{opp}}{\text{adj}}$$

$$A = \frac{1}{2}ab \sin C$$

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

$$c^2 = a^2 + b^2 - 2ab \cos C$$

$$\cos C = \frac{a^2 + b^2 - c^2}{2ab}$$

Financial Mathematics

$$FV = PV(1 + r)^n$$

Straight-line method of depreciation

$$S = V_0 - Dn$$

Declining-balance method of depreciation

$$S = V_0(1 - r)^n$$

Statistical Analysis

An outlier is a score

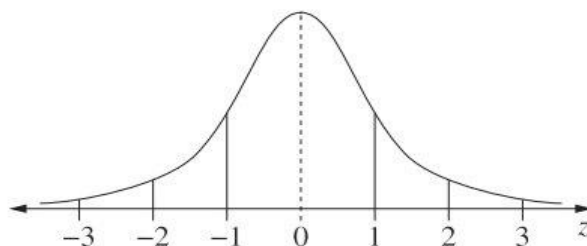
less than $Q_1 - 1.5 \times IQR$

or

more than $Q_3 + 1.5 \times IQR$

$$z = \frac{x - \bar{x}}{s}$$

Normal distribution



- approximately 68% of scores have z -scores between -1 and 1
- approximately 95% of scores have z -scores between -2 and 2
- approximately 99.7% of scores have z -scores between -3 and 3